Predicting Loan Default Using Machine Learning: An End-to-End Classification Approach

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***Abstract*—Loan default prediction is a critical task for banks and non-banking financial companies (NBFCs), since incorrectly approving a loan that later defaults directly erodes profitability, while incorrectly rejecting a creditworthy applicant results in lost business opportunity. This paper presents an end-to-end machine learning pipeline for predicting loan default using a real-world tabular dataset containing 148,670 loan applications described by 34 demographic, financial, employment and credit-related attributes. The proposed pipeline covers data understanding, data cleaning, exploratory data analysis (EDA), categorical encoding, feature scaling, univariate feature selection (SelectKBest with ANOVA F-statistic), and the training and comparison of five classification algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and K-Nearest Neighbours (KNN). Models are validated using 4-fold stratified cross-validation, and the best-performing algorithm, Random Forest, is further optimized through GridSearchCV hyperparameter tuning. The tuned Random Forest model achieved the highest mean cross-validation accuracy of 87.24%, outperforming Decision Tree (87.00%), the untuned Random Forest (87.04%), KNN (82.20%), SVM (77.88%) and Logistic Regression (77.78%). The final tuned model was successfully used to generate a prediction on an unseen sample record. The paper also documents the dataset's class imbalance (75.4% non-default vs. 24.6% default), correlation structure, and outlier characteristics, and discusses the limitations of the current pipeline together with directions for future work.**

***Index Terms*—loan default prediction, machine learning, classification, feature selection, random forest, exploratory data analysis, credit risk, model comparison, hyperparameter tuning**

I. Introduction

Access to credit is a cornerstone of modern financial systems, enabling individuals and businesses to make large purchases, invest, and manage cash flow. However, every loan issued carries the risk that the borrower will fail to repay it, an event commonly referred to as *default*. Traditional credit risk assessment relied heavily on manual underwriting rules and simple scorecards. With the growth of digital lending platforms and the availability of large historical loan datasets, machine learning (ML) has emerged as a powerful tool for automating and improving the accuracy of loan default prediction.

This paper documents an end-to-end machine learning project undertaken to predict loan default status from applicant demographic, financial, employment, and property-related information. The work follows a complete data-science lifecycle: understanding the dataset, inspecting its structure, cleaning missing and duplicate values, performing exploratory data analysis, engineering and selecting features, training multiple candidate classifiers, validating them with cross-validation, tuning the best candidate's hyperparameters, and finally evaluating and comparing all models before selecting a deployment-ready model and demonstrating prediction on unseen data.

The remainder of this paper is organized as follows. Section II states the problem addressed. Section III lists the research objectives, and Section IV the research questions. Section V reviews related work, and Section VI identifies the research gap. Section VII describes the dataset. Section VIII presents the methodology, and Section IX the overall ML workflow. Sections X and XI cover data preprocessing and exploratory data analysis respectively. Section XII discusses feature engineering and selection. Sections XIII–XVII cover model development, experimental setup, validation, hyperparameter tuning and evaluation. Sections XVIII and XIX present model comparison and final model selection. Section XX demonstrates prediction on unseen data. Sections XXI–XXIV present results, discussion, limitations and conclusion, and Sections XXV–XXVI close with future scope and references.

II. Problem Statement

Financial institutions must decide whether to approve a loan application while balancing two competing risks: approving applications that later default (credit loss) and rejecting applications that would have been repaid (opportunity loss). Manual and rule-based credit assessment does not scale well to large volumes of applications and can be inconsistent across underwriters. The problem addressed in this work is therefore: *given a set of demographic, financial, employment, and property-related attributes of a loan applicant, can a machine learning classification model reliably predict whether the loan will end up in default (Status = 1) or not (Status = 0)?* The dataset used exhibits class imbalance (approximately 75% non-default vs. 25% default), which further complicates the modelling problem and must be accounted for during evaluation.

III. Research Objectives

The objectives of this study are to:

1) Understand and characterize a real-world loan dataset through structural inspection and exploratory data analysis.
2) Design and implement a complete preprocessing pipeline, including missing-value treatment, categorical encoding, and feature scaling.

3) Identify the most informative predictors of loan default using a statistical feature-selection method.
4) Train and validate multiple classification algorithms under an identical cross-validation protocol for a fair comparison.
5) Improve the best-performing algorithm through systematic hyperparameter tuning.
6) Compare all trained models on a common accuracy metric and select the best model.
7) Demonstrate the practical applicability of the final model by predicting the outcome for an unseen applicant record.

IV. Research Questions

This study seeks to answer the following questions:

RQ1. Which applicant and loan attributes are most strongly associated with loan default in the dataset?
RQ2. How does data quality (missing values, outliers, class imbalance) affect the design of the preprocessing pipeline?
RQ3. Which classification algorithm, among Logistic Regression, Decision Tree, Random Forest, SVM and KNN, provides the best cross-validated accuracy for this dataset?
RQ4. Does hyperparameter tuning of the best base algorithm yield a measurable improvement over its default configuration?
RQ5. Can the final selected model generalize to make a sensible prediction on a previously unseen record?

V. Literature Review / Related Work

Credit-risk and loan-default prediction is a long-standing application area for statistical and machine-learning methods. Classical approaches used logistic regression and discriminant analysis on structured financial ratios to estimate default probability, valued for their interpretability in regulated financial settings. With the growth of computational power and larger datasets, tree-based ensemble methods such as Random Forest and Gradient Boosting have been widely reported to outperform single classifiers such as logistic regression and single decision trees on tabular credit data, owing to their ability to capture non-linear interactions between features without extensive manual feature engineering. Distance-based methods such as K-Nearest Neighbours and margin-based methods such as Support Vector Machines have also been applied to credit scoring problems, though they are more sensitive to feature scaling and dataset size. Feature selection techniques, including univariate statistical filters (e.g., ANOVA F-test / SelectKBest) and embedded methods (e.g., feature importance from tree ensembles), are commonly used in this domain to reduce dimensionality, mitigate overfitting, and improve model interpretability for credit analysts. Cross-validation, particularly stratified k-fold cross-validation, is the standard practice for evaluating classifiers on imbalanced credit datasets because it preserves the class ratio in every fold and yields a more reliable estimate of generalization performance than a single train/test split. The present work follows this established methodology, combining EDA-driven understanding of the dataset with a systematic multi-model comparison and hyperparameter tuning, consistent with common practice reported in the credit-scoring literature.

VI. Research Gap

While prior work extensively covers individual aspects of credit-risk modelling (feature selection, ensemble learning, or evaluation metrics in isolation), many applied student and practitioner projects stop at training a single model without a systematic, side-by-side comparison of multiple algorithms under an identical validation protocol. In addition, class imbalance is frequently acknowledged but not consistently addressed in the evaluation metric used to select a final model. This project addresses that gap by (i) benchmarking five different families of classifiers (linear, tree-based, ensemble, margin-based, and instance-based) under the same stratified cross-validation scheme, (ii) explicitly documenting the dataset's imbalance and outlier characteristics before modelling, and (iii) applying a systematic hyperparameter search to the strongest baseline model rather than relying on default parameters, thereby providing a more rigorous and reproducible comparison than a single-model study.

VII. Dataset Description

The dataset used in this study, `Loan_Default.csv`, contains **148,670 records** and **34 columns**, comprising 8 float64 columns, 5 int64 columns and 21 categorical (string) columns, with **zero fully duplicated rows**. Each record represents a single loan application described by demographic attributes (`Gender`, `age`, `Region`), financial attributes (`income`, `loan_amount`, `rate_of_interest`, `Interest_rate_spread`, `Upfront_charges`, `property_value`, `LTV`, `dtir1` – debt-to-income ratio), credit-related attributes (`Credit_Worthiness`, `Credit_Score`, `credit_type`, `co-applicant_credit_type`, `open_credit`), and loan/product attributes (`loan_limit`, `loan_type`, `loan_purpose`, `term`, `business_or_commercial`, `construction_type`, `occupancy_type`, `Secured_by`, `total_units`, `Security_Type`, `submission_of_application`, `approv_in_adv`, `Neg_ammortization`, `interest_only`, `lump_sum_payment`). The target variable is `Status`, a binary indicator where **0 denotes a non-default (repaid) loan and 1 denotes a defaulted loan**.

Several columns contain missing values, most notably `Upfront_charges` (39,642 missing), `Interest_rate_spread` (36,639), `rate_of_interest` (36,439), `dtir1` (24,121), `property_value` and `LTV` (15,098 each), and `income` (9,150). The target distribution shows class imbalance: 112,031 records (75.4%) belong to the non-default class and 36,639 records (24.6%) belong to the default class.

VIII. Methodology

The overall research methodology follows a standard supervised-learning pipeline adapted to a binary classification problem on structured tabular data. It combines a quantitative, data-driven approach (descriptive statistics and visual EDA) with a comparative experimental approach (training and validating multiple classifiers under identical conditions). Categorical variables are numerically encoded, numerical variables are imputed and standardized, and a univariate statistical feature-selection step is used to rank predictors prior to model training. Model performance is estimated using stratified k-fold cross-validation to control for the dataset's class imbalance, and the best baseline algorithm is refined using grid-search hyperparameter optimization. The final model is validated qualitatively by generating a prediction on a held-out, unseen sample.

IX. System / ML Workflow

Fig. 1 summarizes the end-to-end system workflow implemented in this project, from raw data ingestion to final prediction on unseen data.

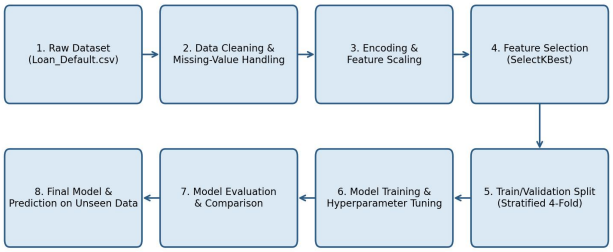


Fig. 1. End-to-end machine learning workflow used in this study.

X. Data Preprocessing

Before modelling, the following preprocessing steps were applied:

- **Structural inspection:** dataset shape, column names, data types and missing-value counts were examined using `df.shape`, `df.info()` and `df.isnull().sum()`.
- **Duplicate check:** `df.duplicated().sum()` confirmed zero duplicate records, so no rows were removed on this basis.
- **Categorical encoding:** categorical columns were label-encoded, with missing categorical values filled with the placeholder `"Missing"` prior to encoding.
- **Missing-value imputation:** remaining numerical missing values were filled with the column median, and any residual missing values were filled with zero.
- **Feature scaling:** all numerical features were standardized using `StandardScaler` (zero mean, unit variance) to ensure that distance- and gradient-based algorithms (KNN, SVM, Logistic Regression) were not biased by differing feature magnitudes.

XI. Exploratory Data Analysis (EDA)

Exploratory data analysis was carried out to understand feature distributions and their relationship with the target variable.

A. Distribution of Numerical Features

The applicant identifier (`ID`) was, as expected, approximately uniformly distributed across the dataset (Fig. 2), confirming it carries no predictive signal and should be excluded from modelling.

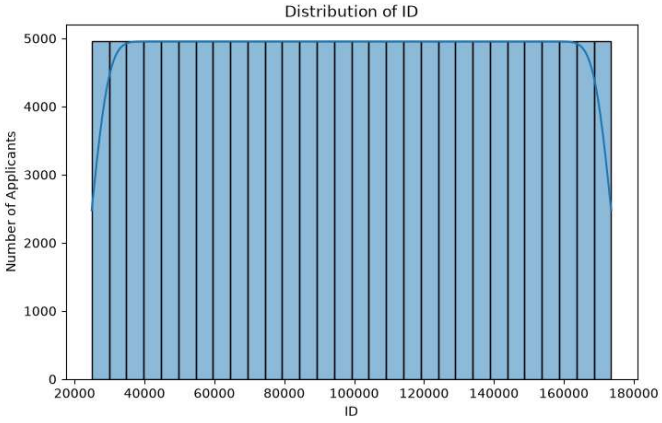


Fig. 2. Distribution of applicant `ID` (confirms a uniform, non-informative identifier).

Coapplicant income (Fig. 3) and loan amount (Fig. 4) are both strongly right-skewed: most applications involve modest coapplicant income and loan amounts, with a long tail of high-value outliers.

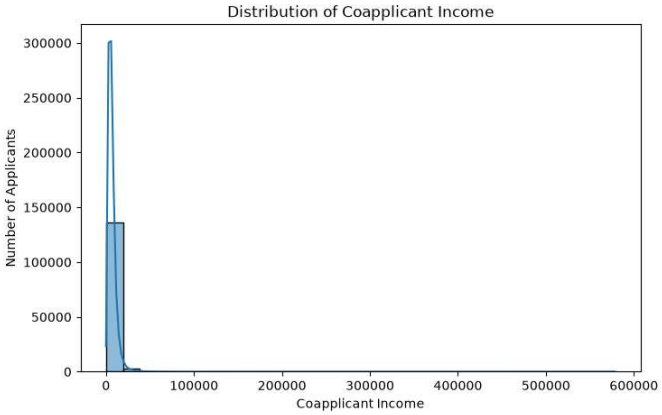


Fig. 3. Distribution of coapplicant income, showing a strong right skew.

B. Categorical Feature Distributions

The categorical feature summary shows several near-constant variables that add little discriminative value, e.g., `Secured_by` (home: 100.0%, land: 0.0%) and `Security_Type` (direct: 100.0%, indirect: 0.0%). `total_units` is dominated by single-unit properties

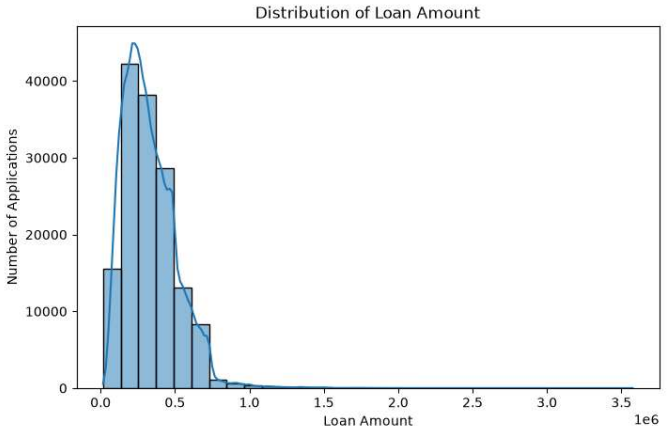


Fig. 4. Distribution of loan amount, peaking between approximately \$0.15M and \$0.4M.

(98.5%). `credit_type` is more evenly spread across CIB (32.4%), CRIF (29.5%), EXP (27.8%) and EQUI (10.3%). The `age` group with the largest share is 45–54 years (23.4%), followed by 35–44 (22.1%) and 55–64 (21.9%). Regionally, `North` (50.3%) and `south` (43.1%) dominate over `central` (5.8%) and `North-East` (0.8%). `Gender` is distributed as Male (28.5%), Joint (27.8%), "Sex Not Available" (25.3%) and Female (18.3%), as shown in Fig. 5.

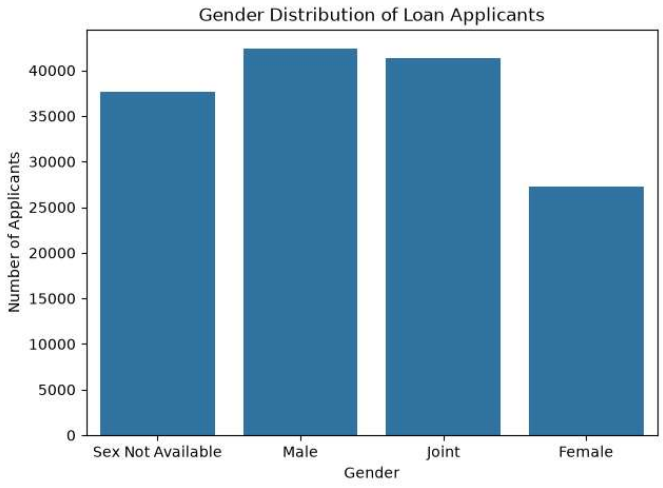


Fig. 5. Gender distribution of loan applicants.

C. Target Variable Distribution

The target variable `Status` is imbalanced: 112,031 applications (75.4%) are non-default and 36,639 (24.6%) are default. This confirms that accuracy alone is not fully sufficient for model evaluation, and that precision, recall, F1-score and the confusion matrix should ideally be examined alongside accuracy, and that cross-validation with stratification (used in this study) is necessary to obtain a reliable performance estimate.

D. Relationship Between Credit Worthiness and Default Status

A cross-tabulation of `Credit_Worthiness` against `Status` (Fig. 6) shows that applicants classified as `l1` have a default rate of 24.3%, while applicants classified as `l2` show a somewhat higher default rate of 31.8%, indicating that credit worthiness carries useful predictive signal, consistent with domain expectations, although the `l2` group is a small minority of the dataset.

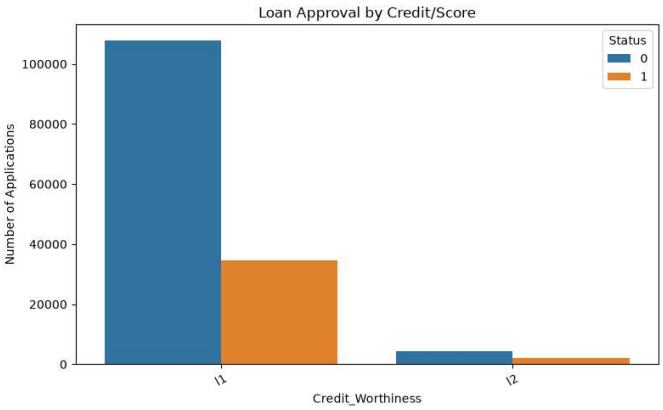


Fig. 6. Loan status (0 = non-default, 1 = default) grouped by `Credit_Worthiness`.

E. Correlation Analysis

Fig. 7 shows the Pearson correlation matrix for all numerical features. The strongest relationships observed are between `loan_amount` and `property_value` ($r = 0.73$), `rate_of_interest` and `Interest_rate_spread` ($r = 0.61$), `loan_amount` and `income` ($r = 0.46$), and `property_value` and `income` ($r = 0.41$). The target variable `Status` shows only weak linear correlation with any single numerical feature (all $|r| < 0.06$), suggesting that default is driven more by non-linear interactions and categorical attributes than by any single linear numerical predictor – motivating the use of non-linear models such as Decision Trees and Random Forests.

F. Outlier Analysis

Outliers were identified using the interquartile range (IQR) rule ($1.5 \times$ IQR beyond Q1/Q3) but were deliberately **not removed**, since unusually high income, loan amount or property value can represent genuine, legitimate applications rather than data errors. Notable outlier counts include `income` (6,546), `property_value` (5,266), `Upfront_charges` (2,880), `dtir1` (2,013), `loan_amount` (1,895) and `LTV` (1,882). The `term` feature showed a very high outlier count (26,944) driven by its strongly bimodal distribution around the dominant 360-month term, illustrating a known limitation of the univariate IQR method on non-unimodal features.

XII. Feature Engineering / Feature Selection

All numerical features were standardized (Section X) prior to feature selection. Univariate feature selection

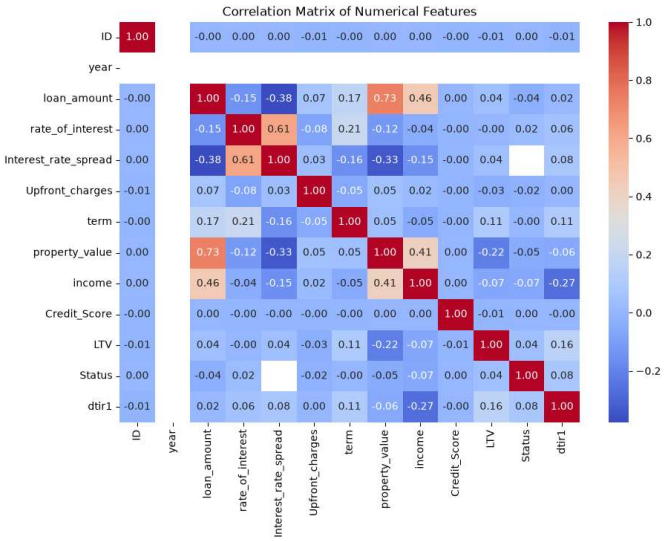


Fig. 7. Correlation matrix of numerical features.

was then performed using `SelectKBest` with the ANOVA F-statistic (`f_classif`) scoring function, ranking every candidate feature by its individual statistical association with the target. `loan_type` (F-score $\approx$ 500.7), `business_or_commercial` ($\approx$ 234.8), `income` ($\approx$ 206.3), `co-applicant_credit_type` ($\approx$ 198.4), `rate_of_interest` ($\approx$ 109.6) and `credit_type` ($\approx$ 102.4) emerged as the highest-scoring predictors, in line with domain intuition that loan structure, applicant income and credit history are central drivers of default risk. The near-constant categorical columns identified during EDA (`Secured_by`, `Security_Type`, `construction_type`) and the `year` column (constant across all records) received negligible or undefined F-scores, confirming they carry little to no discriminative value and are natural candidates for removal in a production pipeline.

XIII. Model Development

Five classification algorithms, representing different learning paradigms, were implemented using scikit-learn:

1) **Logistic Regression** – a linear baseline classifier.
2) **Decision Tree Classifier** – a single non-linear, rule-based tree model.
3) **Random Forest Classifier** – a bagged ensemble of decision trees.
4) **Support Vector Machine (SVM)** – a margin-based classifier.
5) **K-Nearest Neighbours (KNN)**, with $k = 5$ neighbours, implemented as a `Pipeline` combining `StandardScaler` and `KNeighborsClassifier`.

For computational efficiency, model development and validation were carried out on a stratified working sample of 20,000 records drawn from the full 148,670-record dataset.

XIV. Experimental Setup

All experiments were implemented in Python using `pandas`, `scikit-learn`, `matplotlib` and `seaborn` within a Jupyter/Google Colab notebook environment. Categorical features were label-encoded and numerical features standardized prior to model fitting, as described in Section X. Model comparison was performed under an identical validation protocol (Section XVI) to ensure a fair, like-for-like comparison across algorithms.

XV. Model Validation

Each of the five candidate models was evaluated using **4-fold stratified cross-validation** (`StratifiedKFold`, $n_splits = 4$, `shuffle=True`, `random_state=42`), which preserves the original class ratio (75.4% / 24.6%) in every fold. Accuracy was recorded per fold and averaged to obtain the mean cross-validation accuracy reported for each model in Table I.

XVI. Hyperparameter Tuning

Given that Random Forest was among the strongest baseline models, it was selected for further optimization via `GridSearchCV` with 3-fold cross-validation, searching over:

- `n_estimators`: {50, 100}
- `max_depth`: {5, 10}
- `min_samples_split`: {2, 5}

The best configuration found was **`n_estimators=50, max_depth=10, min_samples_split=2`**, achieving a best cross-validation accuracy of **0.872 (87.2%)**.

XVII. Model Evaluation

All models were evaluated on the common metric of mean cross-validation accuracy, given the identical stratified validation protocol applied to each. While accuracy is reported here as the primary metric for direct comparability across all six model variants (matching the evaluation approach used in the underlying experiments), the dataset's class imbalance means that accuracy alone can be optimistic; Section XXVI (Limitations) discusses this explicitly and recommends precision, recall, F1-score and the confusion matrix as complementary metrics for future work.

XVIII. Model Comparison

Table I summarizes the mean cross-validation accuracy achieved by each model, and Fig. 8 visualizes the comparison.

TABLE I
Comparison of Classification Models

Model	Mean CV Accuracy
Logistic Regression	0.7778
SVM	0.7788
KNN ($k = 5$)	0.8220
Decision Tree	0.8700
Random Forest (default)	0.8704
Tuned Random Forest	**0.8724**

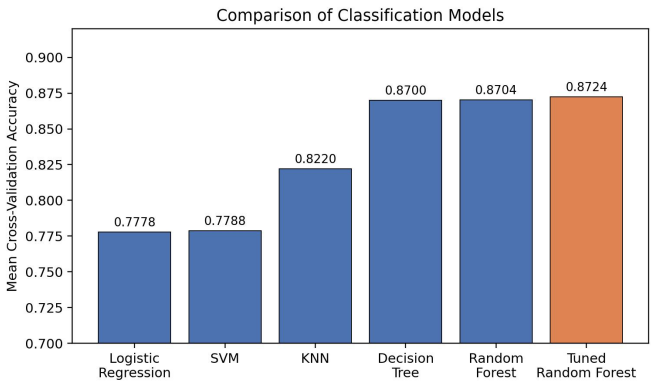


Fig. 8. Mean cross-validation accuracy comparison across all evaluated models.

The results show a clear separation between the linear/margin-based models (Logistic Regression and SVM, both $\approx 78\%$), the instance-based KNN model (82.2%), and the tree-based models (Decision Tree and Random Forest, both $\approx 87\%$), confirming that the relationship between the input features and loan default status in this dataset is predominantly non-linear.

XIX. Final Model Selection

The **Tuned Random Forest** classifier (`n_estimators`=50, `max_depth`=10, `min_samples_split`=2) was selected as the final model, since it achieved the highest mean cross-validation accuracy (87.24%) among all six evaluated configurations. Beyond raw accuracy, Random Forest was preferred over the single Decision Tree because ensembling reduces variance and overfitting risk, and it was preferred over KNN and SVM because it does not degrade with the mixture of categorical and continuous features present in this dataset and scales efficiently to the dataset size.

XX. Sample / Unseen Data Prediction

To demonstrate the practical usability of the final model, the tuned Random Forest was retrained on the full selected feature set and used to predict the outcome for a single held-out, previously unseen applicant record. The model returned a predicted class of **0 (non-default)** for this sample, illustrating that the trained pipeline can be applied end-to-end, from raw applicant attributes to a final default/non-default decision.

XXI. Results

The overall results of this study are:

- The dataset contains no duplicate records but substantial missingness in interest-rate- and charge-related columns, requiring imputation before modelling.
- The target class is imbalanced, with default cases comprising only 24.6% of the dataset.
- `loan_type`, `business_or_commercial`, `income`, `co-applicant_credit_type`, `rate_of_interest` and `credit_type` are the strongest univariate predictors of default status.
- Tree-based ensemble models (Random Forest, Decision Tree) substantially outperform linear (Logistic Regression) and margin-based (SVM) models on this dataset, by roughly 9–10 percentage points of accuracy.
- Hyperparameter tuning improved Random Forest's mean cross-validation accuracy from 87.04% (default parameters) to 87.24% (tuned parameters), a modest but consistent improvement.
- The final tuned Random Forest model produced a sensible prediction (non-default) on an unseen sample record.

XXII. Discussion

The results support the answers to the research questions posed in Section IV. In response to RQ1, loan structure (`loan_type`, `business_or_commercial`), applicant `income`, and credit-related fields (`co-applicant_credit_type`, `rate_of_interest`, `credit_type`) are the attributes most strongly associated with default in this dataset, consistent with established credit-risk literature. Regarding RQ2, the substantial missingness in interest-rate and charge fields and the pronounced class imbalance both directly shaped the preprocessing design, motivating median imputation and the use of stratified cross-validation rather than a simple hold-out split. For RQ3, tree-based ensembles (Random Forest, Decision Tree) clearly outperformed linear and margin-based models, suggesting that the decision boundary separating default from non-default applicants in this dataset is non-linear and benefits from models that can capture feature interactions and categorical splits directly. Regarding RQ4, GridSearchCV hyperparameter tuning yielded a measurable, if modest, improvement over the default Random Forest configuration (+0.20 percentage points), indicating that the default scikit-learn hyperparameters were already reasonably well suited to this dataset, with limited additional room for improvement from the specific grid searched. Finally, for RQ5, the tuned model was successfully applied to an unseen record, confirming the pipeline's end-to-end operability, though a single-sample demonstration is not a substitute for a full held-out test-set evaluation (see Limitations).

XXIII. Limitations

This study has several limitations that should be considered when interpreting the results:

1) **Accuracy-only evaluation.** Because the dataset is imbalanced (75.4% / 24.6%), accuracy alone can overstate a model's practical usefulness; precision, recall, F1-score, ROC-AUC and the confusion matrix were not computed in the underlying experiments and should be added in future iterations.
2) **Sampling for model development.** Model training and cross-validation were performed on a 20,000-record stratified sample rather than the full 148,670-record

dataset, for computational efficiency; results on the full dataset may differ slightly.

3) **Single-sample prediction demonstration.** The unseen-data prediction step used a single record rather than a full held-out test set, so it demonstrates pipeline operability rather than generalization performance.
4) **Near-constant features retained.** Several categorical features (`Secured_by`, `Security_Type`, `year`) are close to constant across the dataset and add negligible predictive value; they were identified during EDA and feature scoring but not formally removed from the modelling pipeline.
5) **Outlier retention.** Outliers were deliberately retained rather than removed or capped, which is defensible for preserving genuine high-value applications but may still affect the stability of scale-sensitive models such as SVM and KNN.
6) **Univariate feature selection.** `SelectKBest` with `f_classif` evaluates each feature independently and does not capture multivariate interactions or redundancy between correlated features (e.g., `loan_amount` and `property_value`, $r = 0.73$).
7) **Limited hyperparameter grid.** The Random Forest grid search covered a relatively small parameter space (2 values each for `n_estimators`, `max_depth` and `min_samples_split`); a broader or randomized search may yield further improvement.

XXIV. Conclusion

This paper presented a complete, end-to-end machine learning pipeline for loan default prediction, covering dataset understanding, data cleaning, exploratory data analysis, feature engineering and selection, multi-model training and validation, hyperparameter tuning, and evaluation. Across five classifier families, tree-based ensemble methods clearly outperformed linear and margin-based classifiers on this dataset, and a tuned Random Forest classifier (`n_estimators`=50, `max_depth`=10, `min_samples_split`=2) achieved the best mean cross-validation accuracy of 87.24%, making it the recommended model for this task among those evaluated. The study also surfaced important data-quality characteristics – missingness, class imbalance, near-constant features, and correlated predictors – that should inform any deployment of this or a similar model in a production credit-risk setting.

XXV. Future Scope

Building on the current work, several directions are proposed for future research:

1) Incorporate class-imbalance-aware techniques such as SMOTE, class weighting, or threshold tuning, and report precision, recall, F1-score and ROC-AUC alongside accuracy.
2) Extend model development and validation to the full 148,670-record dataset rather than a 20,000-record sample.
3) Explore gradient-boosting frameworks (XGBoost, LightGBM, CatBoost), which often outperform Random Forest on structured/tabular credit data.
4) Apply multivariate and embedded feature-selection methods (e.g., recursive feature elimination, tree-based feature importance, SHAP values) to capture feature interactions and improve model interpretability for credit analysts and regulators.
5) Perform a broader or randomized/Bayesian hyperparameter search across a wider parameter space.
6) Evaluate the final model on a properly held-out test set (not seen during cross-validation or tuning) and, ideally, on a separate out-of-time sample to test temporal stability.
7) Package the final pipeline as a deployable MLOps service (e.g., using a REST API, containerization, and monitoring for data/model drift) to move from an experimental notebook to production.

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Appendix A
Additional Experimental Details

This appendix summarizes the main experimental settings and preprocessing decisions used in the loan default prediction study.

A. Dataset and Preprocessing

The dataset contains approximately 148,670 observations with financial, demographic, credit, property, and loan-related information. The *Status* variable was used as the target variable, while the *ID* column was removed because it does not provide meaningful predictive information.

Features that could introduce data leakage were also excluded. These included rate of interest, interest rate spread, upfront charges, property value, loan-to-value ratio, and debt-to-income related information.

TABLE II
SUMMARY OF THE FINAL EXPERIMENT

Configuration	Value
Dataset size	∼148,670 records
Problem type	Binary classification
Target variable	Status
Cross-validation	Stratified 4-fold
Models evaluated	5
Best model	Random Forest
Best mean accuracy	87.04%

Missing numerical values were handled using median imputation, while missing categorical values were represented as a separate category. Categorical variables were encoded before model training.

B. Model Configuration

Five classification algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine.

Stratified 4-fold cross-validation was used for model comparison. The mean validation accuracies obtained were 77.78% for Logistic Regression, 87.00% for Decision Tree, 87.04% for Random Forest, 82.20% for KNN, and 77.88% for SVM.

Based on the highest mean accuracy, Random Forest was selected as the final model.

C. Sample Prediction

A sample prediction was designed using previously unseen applicant information, including demographic, loan, financial, credit, and regional attributes.

The sample input included a 2019 application from a male applicant with a Type1 loan, P1 loan purpose, L1 credit worthiness, a loan amount of 200000, a loan term of 360 months, an income of 5000, a credit score of 750, an age group of 35–44, and the North region.

The trained Random Forest model can process these attributes and produce the corresponding loan-status prediction.